

ContextualGraph-LLM: A multimodal framework for enhanced Darknet traffic analysis

Yujung Hwang^a, Furkan Kurt^a, Faruk Curebal^b, Omer Keskin^b, Abdulhamit Subasi^{a,*}

^a Information Sciences & Technology Dept., College of Emergency Preparedness, Homeland Security & Cybersecurity, University at Albany, SUNY, Albany, NY, USA

^b Cybersecurity Dept., College of Emergency Preparedness, Homeland Security & Cybersecurity, University at Albany, SUNY, Albany, NY, USA

ARTICLE INFO

Keywords:

Intrusion detection
Darknet Traffic
Graph-Based Machine Learning
Lightweight Language Models
ContextualGraph-LLM

ABSTRACT

Darknet environments present a dual challenge for cybersecurity, and they offer privacy for legitimate users yet also serve as a hub for illicit activities. Traditional detection methods struggle to cope with the sophisticated tactics employed by threat actors in this space, necessitating innovative solutions that can uncover nuanced malicious behaviors. The primary objective of this work is to overcome the limitations of traditional detection methods against nuanced malicious behaviors in Darknet environments by introducing ContextualGraph-LLM (CG-LLM). This novel framework leverages the combined power of Graph Neural Networks (GNNs) and Large Language Models (LLMs) to enhance multi-label intrusion detection accuracy across various real-world network traffic scenarios, as demonstrated by its superior performance on benchmark datasets. This paper proposes CG-LLM, a novel framework that integrates Graph Neural Networks (GNNs) with Large Language Models (LLMs) to achieve robust multi-label intrusion detection. Our approach capitalizes on the CIC-Darknet2020 dataset, along with the ToN_IoT and UNSW-NB15 datasets, which cover a variety of real-world scenarios reflective of actual Darknet traffic and broader network threats. We first transform raw network flow records into a bipartite graph structure, where nodes correspond to distinct endpoints (source–destination pairs) and edges capture flow-level features such as packet sizes and transaction volumes. These graph-based embeddings are then enriched with textual node embeddings derived from TinyLlama. By pairing structural insights with contextual semantics, CG-LLM is poised to detect subtle malicious patterns often overlooked by more conventional, single-modality methods. Our experimental results reveal that CG-LLM consistently surpasses state of the arts studies in weighted average F1-score (0.9035), precision, (0.9368) and recall (0.8803) on the CIC-Darknet2020 dataset. Furthermore, it achieved impressive results on ToN_IoT (F1-score: 0.9712, precision: 0.9750, recall: 0.9714) and UNSW-NB15 (F1-score: 0.9797, precision: 0.9809, recall: 0.9802), highlighting its capacity to tackle data imbalance and underrepresented attack categories. Notably, the model excels in detecting hard-to-spot malicious behaviors, underscoring the effectiveness of fusing graph topology with linguistic context. While the incorporation of LLM embeddings elevates detection accuracy, it also introduces computational overhead, pointing to the importance of future research on scalability and optimization. In conclusion, this study demonstrates the potential of multimodal architectures that harness the complementary strengths of GNNs and LLMs, offering an adaptable and resilient framework for safeguarding networks against ever-evolving Darknet threats.

1. Introduction

The Dark web serves as both a refuge for preserving privacy and a breeding ground for illicit activities, accessible only through specialized tools like the Tor browser. This segment of the Internet (Fig. 1), which ensures anonymity by encrypting and rerouting user traffic (Ablon & Libicki, 2015) across multiple nodes, poses unique challenges for

cybersecurity professionals, particularly due to its association with both legitimate and illegal activities (Mahboubi et al., 2024; Tazi et al., 2022). The anonymization that tools like Tor network provide, while critical for many users, complicates efforts to monitor and mitigate potential cyber threats (Safaei Pour et al., 2020).

Dark web marketplaces and forums play a pivotal role in facilitating cybercriminal activities (Christin, 2013), serving as platforms for

* Corresponding author.

E-mail addresses: yhwang5@albany.edu (Y. Hwang), fkurt@albany.edu (F. Kurt), fcurebal@albany.edu (F. Curebal), okeskin@albany.edu (O. Keskin), asubasi@albany.edu (A. Subasi).

<https://doi.org/10.1016/j.eswa.2025.129298>

Received 12 January 2025; Received in revised form 24 July 2025; Accepted 4 August 2025

Available online 6 August 2025

0957-4174/Published by Elsevier Ltd.

trading stolen credentials, exploit kits, and other malicious tools. According to a cyber threat intelligence report (Searchlight [Cyber and Marsh McLennan, 2024](#)), the presence of darknet traffic in a corporate network leads more than twice the likelihood of a successful breach in the following year. Furthermore, the combined risk of multiple dark web exposures, such as outgoing traffic and OSINT findings, can raise the likelihood of a cyberattack by up to 77 %.

Similarly, the *X-Force Threat Intelligence Index 2024* report by [IBM \(2024\)](#) highlights the increased use of valid credentials as an initial access vector, with cloud credentials making up 90 % of compromised cloud assets sold on the dark web. This trend underscores the critical role of dark web intelligence in pre-attack phases, where attackers leverage these resources to plan sophisticated breaches. The symbiotic relationship between dark web activities and rising cyber threats necessitates proactive measures, such as continuous monitoring of dark web exposure, to mitigate these risks effectively.

The evolution of cyber threats regarding the Darknet traffic in the corporate network necessitates advanced detection methods capable of identifying subtle signs of malicious activity. Traditional cybersecurity measures often struggle to keep pace with the sophisticated techniques employed by attackers leveraging the Darknet. This has led to the adoption of innovative technologies such as Graph Neural Networks (GNNs) and Large Language Models (LLMs), which offer new strategies to enhance network security and intelligence. This study proposes a novel approach for detection of darknet traffic within the internal network.

GNNs have shown exceptional promise in modeling network data to detect irregular patterns and anomalies indicative of cyber threats. Their ability to process and analyze graph-structured data makes them ideal for understanding the complex interactions within network traffic. For instance, studies like those by [Bilot et al. \(2023\)](#) and [Tran & Park \(2024\)](#) have demonstrated how GNNs can effectively enhance intrusion detection systems (IDS) by capturing dependencies and anomalies in network behavior. Parallely, LLMs are being utilized to interpret and analyze unstructured data related to the Darknet traffic, providing deep contextual insights that are crucial for identifying and responding to security threats. The integration of LLMs with GNNs has been explored in several studies, including [Farrukh et al., 2024](#)) and [Liu et al. \(2024\)](#), to create comprehensive models that not only detect but also predict potential security breaches before they occur.

This study employs the CIC-Darknet2020 dataset ([Lashkari et al., 2021](#)) a rich collection of network traffic data curated by the Canadian Institute for Cybersecurity. This dataset includes varied traffic scenarios that mimic real-world network behaviors within Darknet environments, providing a robust platform for testing and enhancing our proposed framework that employs GNN and LLM. To further validate the robustness and generalizability of our framework, we also incorporate two additional prominent datasets: ToN_IoT ([Alsaedi et al., 2020](#); [Ashraf et al., 2021](#); [Booij et al., 2022](#); [Moustafa, 2019a, 2019b, 2021](#); [Moustafa, Ahmed, et al., 2020](#); [Moustafa, Keshky, et al., 2020](#)) and UNSW-NB15 ([Moustafa & Slay, 2015](#)). The use of these specific datasets enables the practical application of theoretical models in a controlled yet realistic setting, facilitating the development of more effective cybersecurity measures. As cyber threats become advanced, the technologies deployed

to counter them must advance accordingly. The integration of GNNs and LLMs into cybersecurity strategies offers a promising path forward, enhancing our ability to monitor, analyze, and respond to threats related to the Darknet. By leveraging comprehensive datasets like CIC-Darknet2020, this research not only tests the efficacy of these advanced technologies but also contributes to the broader understanding of cybersecurity in the age of anonymity.

The novel contribution of this paper is to create a ContextualGraph-LLM (CG-LLM), a novel framework that integrates Graph Neural Networks (GNNs) with Large Language Models (LLMs) to achieve robust multi-label intrusion detection in Darknet traffic. We first transform raw network flow records into a bipartite graph structure, where nodes correspond to distinct endpoints (source–destination pairs) and edges capture flow-level features such as packet sizes and transaction volumes. These graph-based embeddings are then enriched with textual node embeddings derived from a compact yet powerful LLM variant, TinyLlama. By pairing structural insights with contextual semantics, CG-LLM is poised to detect subtle malicious patterns often overlooked by more conventional, single-modality methods.

The rest of the paper is organized as follows: [Section 2](#) provides an in-depth review of related work, focusing on existing graph-based intrusion detection models and their limitations. [Section 3](#) outlines the methodology, describing the proposed model's architecture and key components. [Section 4](#) presents the experimental setup and evaluation results, comparing the proposed model to baseline models using comprehensive metrics and per-class analyses, and discussion of the implications of the findings and provides avenues for future research. Finally, [Section 5](#) concludes the article by summarizing the key points.

2. Literature review

Darknet traffic analysis has emerged as a critical area of research due to the increasing complexity of network security threats. Researchers have employed diverse methodologies, including machine learning (ML), deep learning (DL), graph-based models, and hybrid approaches, to address challenges in traffic detection and anomaly classification. Machine learning has been pivotal in enhancing darknet traffic classification accuracy. [Abu Al-Haija et al. \(2022\)](#) introduced the BAG-DT model, which achieved 99.50 % accuracy using six supervised techniques, outperforming existing models by up to 27 %. [Almomani \(2023\)](#) employed stacking ensemble learning for malignant and benign traffic classification, achieving over 97 % accuracy during testing. Similarly, [Meidan et al. \(2017\)](#) proposed a machine learning approach for the Internet of Things (IoT) device classification, achieving 99.28 % accuracy in distinguishing IoT and non-IoT devices. [Rust-Nguyen et al. \(2023\)](#) assessed multiple machine learning models, finding that Random Forest outperformed other classifiers, achieving an F1-score of 0.998 for traffic classification with the CIC-Darknet2020 dataset and demonstrating resilience against class imbalance through the use of Synthetic Minority Oversampling Technique. Enhancing malware classification has also been a focus; one study assessed seven feature selection algorithms with parameter optimization, demonstrating the necessity of tailored feature selection approaches to optimize model performance across datasets ([Curebal & Dag, 2024](#)).

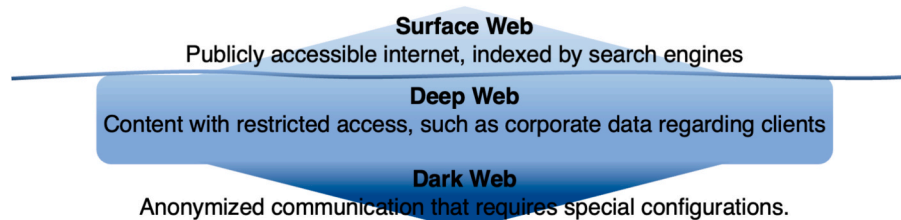


Fig. 1. Dark web and the other segments of the internet.

Deep learning has significantly advanced the field with its ability to model complex traffic patterns. Alimoradi et al. (2022) developed a Tor-VPN detector using deep neural networks, achieving 96 % accuracy without requiring preprocessing. Fatani et al. (2021) proposed an IDS combining Convolutional Neural Networks (CNN) with metaheuristic algorithms, resulting in superior performance compared to traditional methods. Hwang et al. (2020) designed D-PACK, a DL-based system capable of detecting malicious traffic with nearly 100 % accuracy by analyzing just two packets per flow. Singh et al. (2021) utilized transfer learning with VGG19, achieving 96 % accuracy in darknet malware classification.

Graph-based techniques have gained prominence for their ability to capture relational and topological information. Chang and Branco (2024) introduced E-GraphSAGE and E-ResGAT, addressing class imbalance issues in intrusion detection. Deng et al. (2023) proposed FT-GCN, a graph convolutional network designed for IoT intrusion detection, achieving high accuracy with limited labeled data. Li & Chasaki (2024) advanced graph techniques with heterogeneous graph neural networks, introducing Express Edges to optimize attack detection. Zhu et al. (2024) developed DGNN, a graph neural network incorporating multi-head attention for darknet traffic classification, outperforming baseline models.

Hybrid models have demonstrated versatility in addressing multifaceted challenges in darknet analysis. Mohanty et al. (2022) proposed a stacking ensemble model combining decision trees, random forests, and autoencoders, achieving 98.89 % accuracy and demonstrating robustness against adversarial attacks. Pei et al. (2024) introduced NIGNN, a privacy-preserving graph neural network that balances privacy concerns and detection accuracy. Lan et al. (2022) developed DarknetSec, a self-attentive DL model combining CNN and Bi-LSTM architectures, achieving 92.22 % accuracy. A recent study combined novel feature selection algorithms with an ensemble of classifiers, using a correlation matrix to eliminate redundant information and improve malware classification accuracy by 1.2 % on the Microsoft Malware Prediction dataset (Cürebal et al., 2023).

Several studies have focused on frameworks to improve traffic analysis and anomaly detection. Demertzis et al. (2021) proposed a system using weight-agnostic neural networks (WANNs) to automate malicious intent detection and identify zero-day attacks. Wang & Zhu (2022) introduced a graph-based behavioral anomaly detection framework, leveraging co-occurrence properties for enhanced accuracy. Efforts to improve efficiency and interpretability have gained traction. Hu et al. (2023) emphasized the interpretability of Dark-DKGC's classifications through knowledge graphs, enhancing understanding of model predictions. Li & Pan (2020) reviewed deep learning approaches for network traffic classification, highlighting challenges such as dataset availability and scalability.

Cybersecurity challenges in darknet environments remain a pressing concern. Duan et al. (2024) proposed a continuous temporal graph neural network (CTGNN) to detect low-frequency behaviors in 5G/B5G cyberspace, achieving superior performance. Ganapathiyappan & Yogish (2024) employed an attention LSTM model to classify darknet operations with 97 % accuracy, showcasing explainable AI techniques like LIME for model interpretation.

Recently, LLMs have emerged as transformative tools in traffic analysis, offering capabilities beyond traditional ML and DL methods. Li et al. (2024) proposed IDS-Agent, an LLM-based intrusion detection system that integrates external knowledge for explainable predictions, demonstrating a 0.97 F1 score on the ACI-IoT benchmark and identifying zero-day attacks with a recall of 0.61. Chen et al. (2024) presented MERLOT, a scalable mixture-of-experts (MoE) framework leveraging distilled LLMs for encrypted traffic classification. Adjewa et al. (2024) proposed a continuous intrusion detection framework utilizing a transformer encoder architecture, achieving 100 % recall in detecting malicious traffic while maintaining high accuracy. These advancements underscore the versatility of LLMs in integrating external knowledge,

adapting to new attack patterns, and scaling to diverse environments, addressing limitations in traditional methods.

The reviewed literature underscores significant advancements in darknet traffic analysis using machine learning, deep learning, graph-based, and hybrid approaches. These studies highlight the potential of leveraging advanced computational models for anomaly detection, traffic classification, and intrusion detection. By utilizing the contextual understanding and generative capabilities of LLMs, we aim to create a system capable of enhancing traffic classification, providing interpretable insights, and addressing scalability issues in darknet environments. The subsequent sections outline our unique methodology, emphasizing the integration of LLMs with traditional analytical frameworks to push the boundaries of darknet traffic analysis.

3. Methodology

This section introduces the methodology underlying the proposed CG-LLM model. It begins with an overview of the dataset used and the data preprocessing techniques employed to ensure effective input representation. Subsequently, the model's architecture is described, focusing on how contextual embeddings from TinyLlama are integrated with graph-structured data to enhance intrusion detection.

3.1. Graph construction

In this study, we utilize graph-structured data constructed following the methodology proposed by Chang and Branco (2024). The network flow data, consisting of source addresses, destination addresses, and flow features (e.g., duration, transaction bytes, packet sizes), is described by a bipartite graph $G(S, D; E)$. In this graph representation, S denotes the set of source nodes, which are uniquely defined by combining IP addresses and port numbers. Similarly, D indicates the set of destination nodes, constructed in the equivalent manner to ensure a distinct representation of network endpoints. The edges, E , represent the network flows between these nodes, with each edge carrying attributes derived from the flow features, such as flow duration, transaction bytes, and packet sizes. This structured representation enables the modeling of complex network interactions while preserving the contextual and structural relationships inherent in the data.

To support efficient graph learning, we encode each edge with a N -dimensional feature vector, capturing flow metadata such as packet sizes, time intervals, and protocol-specific attributes. The edge attributes are normalized and stored in `edge_feat_scaled.npy`, while the adjacency information for the bipartite graph is stored in `adj_random.npy`. Each edge corresponds to a node pair (source, destination), which is also converted into a natural-language prompt in the format: "What is the relationship between Node A and Node B? Choices: [labels]". These prompts are processed by the LLM to extract semantic embeddings, enabling contextual interpretation of network behavior.

Although our graph representation is static, the edge features incorporate several time-derived attributes from all three datasets. In the CIC-Darknet2020 dataset, we utilized statistical summaries and features such as inter-arrival times (fIAT, bIAT, flowIAT), flow durations, and idle/active periods. Similarly, in the UNSW-NB15 dataset, we extracted both direct time features such as duration, record start and last times, and inter-packet arrival times, and indirect timing-related indicators, such as source and destination bits per second, and TCP setup metrics. For the ToN-IoT Dataset, duration and features like, `src_bytes`, `dst_bytes`, and packet volume attributes were included, which encode temporal dynamics implicitly. Across all datasets, these temporal indicators were embedded as part of the feature vectors, allowing the model to learn from behavioral timing patterns despite the static graph design.

This bipartite graph structure encodes the intrusion detection problem as an edge classification task. Additionally, the graph can be converted into a line graph, where edges in the bipartite graph are displayed

as nodes. This transformation reformulates the problem as a node classification task, enabling the use of advanced graph learning techniques. By leveraging this pre-constructed graph representation, we introduce a novel approach that combines structural insights from graph-based data with contextual information extracted using LLMs, providing an effective framework for enhancing multi-label intrusion detection.

3.2. Model architecture

In this section, we propose a **CG-LLM** model to concatenate node textual embeddings and edge features into a unified framework. The overall architecture of the proposed CG-LLM model is illustrated in Fig. 2. The architecture begins with the construction of a graph from the CIC-Darknet dataset, where nodes represent network entities (e.g., IP addresses and ports) and edges correspond to network flows.

We employ the TinyLlama-1.1B model as the LLM component in our multimodal architecture. The model is fine-tuned to process structured node-pair queries generated from the graph and extract contextual embeddings.

3.2.1. Edge contextualization

In the CG-LLM model, edge contextualization focuses on processing the relationships (edges) between nodes in the graph to capture important structural information. As shown in left path of Fig. 2, the raw edge feature matrices, which have 77 dimensions, are passed through a fully connected linear layer to refine the information. A ReLU activation is applied to introduce non-linearity, followed by a Dropout layer to prevent overfitting. This process reduces the edge feature dimensions to 64, a common practice for improving efficiency and generalization in graph-based models (P. Li et al., 2023). This dimensionality reduction ensures that key structural patterns in the graph are preserved while minimizing redundancy. We use a linear layer (edge_fc) to encode the original 77-dimensional edge vectors into 64-dimensional representations. These encoded edge features are subsequently combined with the LLM-based node embeddings during classification.

3.2.2. Node contextualization

The node contextualization step leverages the TinyLlama model (P. Zhang et al., 2024), a lightweight variant of Llama 2, to process the textual descriptions associated with each node.

First, as illustrated in the Node Textual Information Processing section of Fig. 2, we generate structured natural language descriptions to capture the relationship between a pair of nodes. These descriptions combine attributes from source and destination nodes (e.g., IP addresses and port numbers) with the label indicating the type of network activity. For instance, if the source node corresponds to a host with ID 10.152.152.11:36882, the destination node corresponds to a server with ID 31.13.90.6:443, and the traffic type is "Browsing," the generated sentence might read: "Traffic observed from 10.152.152.11:36882 to 31.13.90.6:443. What is the relationship between them? Choices: {Labels}." This approach allows us to represent structured network flow data in a format that is interpretable by LLMs, capturing both entity and relation-level context.

Next, the generated text undergoes tokenization using TinyLlama's tokenizer, which breaks the sentence into subword units and prepares it for embedding. TinyLlama then encodes the tokenized input into a 2048-dimensional embedding vector. This embedding captures both the individual attributes of each node and their relational context. TinyLlama maintains the same reasoning and language understanding capabilities as Llama 2, while drastically reducing memory usage through its compact architecture with 1.1 billion parameters. Despite its size, TinyLlama achieves efficient inference and strong performance due to pre-training on a large-scale corpus of 3 trillion tokens (P. Zhang et al., 2024). Internally, the TinyLlama architecture employs LlamaMLP modules to expand the embeddings temporarily from 2048 dimensions to 5632 dimensions before projecting them back to 2048 dimensions. This expansion-enrichment-compression step enhances the expressiveness of the embeddings. To integrate with the edge features, we truncate the final LLM output to 64 dimensions. This ensures compatibility with the 64-dimensional edge embeddings obtained during edge contextualization. Finally, we concatenate the LLM-derived 64-dimensional node

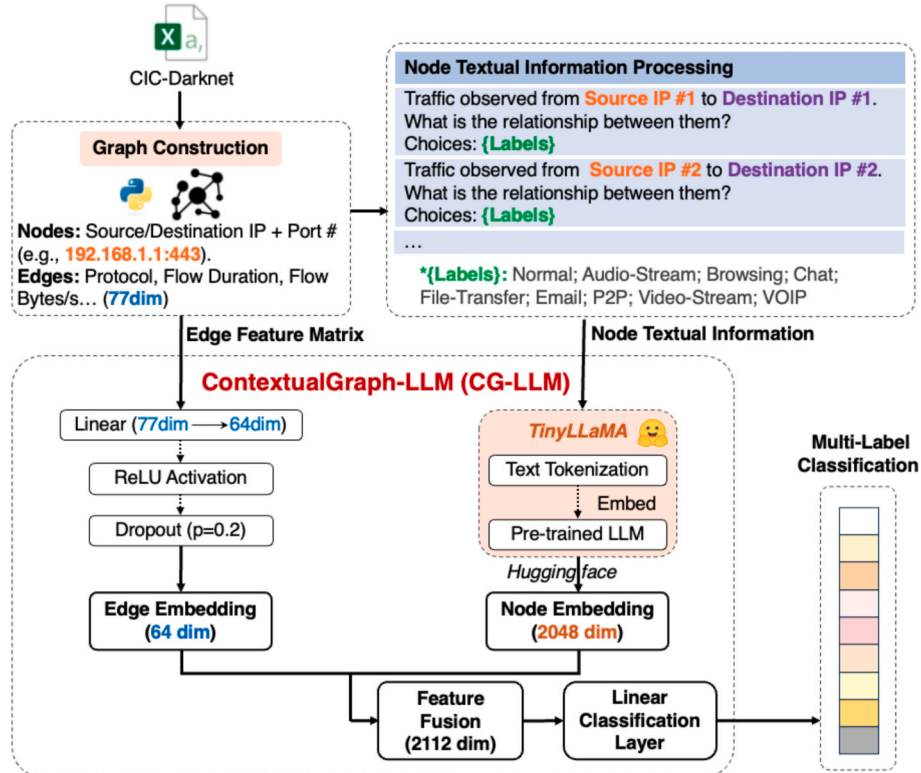


Fig. 2. The proposed architecture of CG-LLM for multi-label intrusion detection.

embeddings with the 64-dimensional edge embeddings and pass the result into a classifier, which outputs one of 9 traffic or attack categories via softmax.

3.2.3. LLM-GNN integration

To enable effective multimodal learning, we integrate the semantic embeddings derived from TinyLlama with the structural features extracted from the graph's edge information. This section provides a formal description of the fusion process and explains the reasoning behind the architectural choices.

Let:

- $E \in \mathbb{R}^{(n \times d_e)}$: the matrix of raw edge feature for n samples, where $d_e = 77$,
- $T \in \mathbb{R}^{(n \times d_t)}$: the matrix of textual embeddings generated from LLM prompts,
- $\varphi_e(\cdot)$: the edge encoder, which projects edge vectors to a lower-dimensional representation via a linear layer,
- $\varphi_t(\cdot)$: the TinyLlama-based encoder, mapping natural language prompts to dense vector representations,
- $\varphi_c(\cdot)$: the classifier operating on the fused representation.

For each sample i_i , the model applies the following transformations:

$$e_i = \varphi_e(E_i), t_i = \varphi_t(prompt_i) \quad (1)$$

$$z_i = [t_i || e_i] \in \mathbb{R}^{128}, \hat{A} \cdot i = \varphi_c(z_i) \quad (2)$$

$$Loss : \mathcal{L} = CrossEntropy(\hat{A}_i, y_i) \quad (3)$$

In this setup, the prompt $prompt_i$ is a structured sentence describing the flow from source to destination node and is designed to elicit relational information from the LLM. As described in Section 3.2.2, the LLM encoder outputs a 2048-dimensional embedding that is truncated to 64 dimensions to match the size of the edge embeddings.

This truncation not only reduces computational cost but also helps mitigate overfitting (Lücke & Eggert, 2010) by discarding overly specific semantic representations that may not generalize well across flows.

The edge features, which capture structural properties such as byte counts, durations, and flags, vary in dimensionality depending on the dataset (e.g., 77 dimensions for CIC-Darknet2020, different dimensions for ToN-IoT or UNSW-NB15).

To standardize these features for fusion, we apply a fully connected transformation φ_e that maps each dataset-specific edge vector $E_i \in \mathbb{R}^{d_e}$ to a shared 64-dimensional latent space. This shared embedding size ensures balance and compatibility during the fusion stage, regardless of dataset heterogeneity.

Unlike traditional GNNs that rely solely on graph topology and numerical features, our approach augments the model's representation space with language-informed embeddings, enabling contextual reasoning over entity relationships. The concatenated feature $e_i \in \mathbb{R}^{128}$ thus, captures both semantic and structural aspects of network flows, leading to improved generalization in multi-class intrusion detection tasks.

This hybrid representation is particularly valuable in detecting attack types that may exhibit similar statistical signatures but differ in their role or intent — distinctions that LLM is well-suited to capture. However, this approach may inherit biases or limitations from the pre-trained LLM. For example, Future work may consider fine-tuning or instruction-tuning the LLM on domain-specific data to prompts phrased in ways that deviate from TinyLlama's training distribution may yield unstable embeddings improve robustness (P. Liu et al., 2023).

3.2.4. Feature fusion and classification

As shown in the feature fusion and classification section, the model

combines the edge embeddings of 64 dimensions and node textual embeddings of 2048 dimensions through a concatenation operation. This results in a unified fused representation with 2112 dimensions, capturing both structural and contextual features of the graph. The fused representation is passed through a fully connected layer to reduce its dimensionality to 128, optimizing it for downstream classification. Finally, the reduced embeddings are processed by a linear classification layer, which outputs predictions for multiple intrusion labels, such as "Normal," "Malicious," "P2P," etc. By combining edge-level structural information and node-level contextual embeddings, the CG-LLM model provides robust and scalable multi-label detection, especially involving high-dimensional and imbalanced intrusion datasets.

4. Results and discussion

This section explains dataset transformation and the experimental setup, including the hyperparameter selection, evaluation metrics, and training strategies, establishing a robust foundation for reproducibility and fair performance comparison. Subsequently, the results of the proposed model are presented and benchmarked with existing state-of-the-art graph-based models.

4.1. Dataset and graph construction

This study employs three publicly available network traffic datasets: CIC-Darknet, ToN-IoT, and UNSW-NB15, each selected for its realism and relevance to intrusion detection. The CIC-Darknet dataset (Habibi Lashkari et al., 2020) contains darknet-related traffic labeled with various attack types such as Botnet, DoS, and Normal, offering packet-level information suitable for flow-based analysis. The ToN-IoT (Alsaedi et al., 2020; Ashraf et al., 2021; Booij et al., 2022; Moustafa, 2019a, 2019b, 2021; Moustafa, Ahmed, et al., 2020; Moustafa, Keshky, et al., 2020) comprises telemetry and network data collected from a realistic IoT/IIoT testbed, with each interaction labeled as normal or a specific cyberattack such as DDoS or ransomware. Only the network portion of this dataset was utilized in this work. The UNSW-NB15 dataset (Moustafa & Slay, 2015) is a well-established benchmark that provides diverse network traffic scenarios, encompassing multiple attack types and normal behavior, and has been widely adopted in intrusion detection research.

To ensure consistency across datasets, a unified graph construction methodology was applied. Each dataset was first parsed from its original CSV format and preprocessed to extract meaningful entities and interaction metadata. Nodes were defined as unique source and destination IP addresses or device identifiers, depending on the dataset. Each traffic record was represented as a directed edge between two nodes, capturing the flow of communication or interaction. Numerical attributes such as protocol information, packet and byte counts, and flag values were extracted from each record and normalized to serve as edge features. In the case of CIC-Darknet, 77 normalized features were used, while ToN-IoT incorporated 39 normalized features per edge, and the UNSW-NB15 dataset utilized 43 edge features. Each edge was assigned a multi-class label based on the original "Label" column, indicating the type of network activity or attack. To support graph-based learning, both an adjacency matrix and an adjacency list were generated to efficiently represent node connections and facilitate model processing.

The final output was formatted into a graph structure compatible with GNN frameworks and CG-LLM model. This structure included nodes, edges, their respective features, and labels, enabling the application of machine learning models for tasks such as classification and link prediction. By transforming the raw tabular data into a graph representation, the dataset effectively captures the inherent relationships and interactions present in network traffic. The dataset preparation process was inspired by methods outlined in the work of Lo et al. (2022), which demonstrated effective approaches to preparing graph-structured data for intrusion detection tasks.

4.2. Performance evaluation metrics

To evaluate the performance of the proposed model, multiple metrics were employed to ensure a thorough assessment of its capabilities. The weighted average F1-score was used as the primary metric to measure the balance between precision and recall while accounting for class imbalances across the dataset. This provides an overall view of the model's ability to classify all categories accurately. In addition, precision (Eq. (4)) and recall (Eq. (5)) metrics were calculated separately to analyze specific aspects of the model's performance. Precision evaluates the accuracy of the model's true positive (TP) predictions by minimizing false positives (FP), while recall assesses its effectiveness in identifying true positives across all classes (Subasi, 2020). Precision evaluates the accuracy of the model's positive predictions by minimizing false positives, while recall assesses its effectiveness in identifying true positives across all classes.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{F1 Score} = \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \quad (6)$$

To gain further insights, per-class F1-scores (Eq. (6)) were used to evaluate the model's performance at the individual class level. This approach helps understand how well the model performs for each category, particularly in scenarios with imbalanced data or challenging classifications. The labels used in this study include 'Normal', 'Audio-Streaming', 'Browsing', 'Chat', 'File-Transfer', 'Email', 'P2P', 'Video-Streaming', and 'VOIP'. These labels represent distinct types of network activities, enabling a detailed evaluation of the model's ability to detect and classify various traffic patterns effectively. These metrics provide a clear and effective way to evaluate how well the model performs across multiple classes, ensuring suitability for intrusion detection. These metrics provide a clear and effective way to evaluate how well the model performs across multiple classes, ensuring suitability for intrusion detection (Kevric et al., 2017; Subasi et al., 2021).

The Area Under the ROC Curve (ROC-AUC) is a key metric for evaluating binary classification models, representing the likelihood that a model ranks a randomly selected positive instance higher than a negative one. AUC values range from 0.5 (random guessing) to 1 (perfect classification). The ROC curve itself plots sensitivity versus 1-specificity across different thresholds, offering a comprehensive view of model performance. A high AUC signifies a good balance between true positives and false positives, making it especially useful for assessing models on imbalanced datasets, where traditional accuracy metrics may be misleading (Bradley, 1997; Hand & Till, 2001).

4.3. Models and hyperparameters

In this study, we benchmark the proposed approach against four established models to evaluate its performance on graph-structured data: 1) E-GraphSage (Lo et al., 2022), 2) GAT (Velickovic et al., 2018), 3) E-ResSage (Chang & Branco, 2024), and 4) E-ResGAT (Chang & Branco, 2024). These models represent advancements in graph-based machine learning, each addressing specific challenges associated with graph representations.

E-GraphSAGE (Lo et al., 2022), an extension of the GraphSAGE framework, incorporates edge feature aggregation to enhance the representation of graph topology and flow characteristics. While it effectively captures edge-level information for edge classification, its reliance on aggregation may underutilize critical edge-specific details, potentially limiting its expressiveness in complex scenarios (Chang & Branco, 2024).

GAT (Velickovic et al., 2018) introduces an attention mechanism to the graph neural network paradigm, enabling adaptive weighting of neighbors during information propagation. This mechanism allows the model to focus on the most influential neighbors, achieving strong performance across various graph-based classification tasks.

E-ResSAGE and E-ResGAT, both proposed by Chang and Branco (2024), introduce residual learning into the graph neural network paradigm. E-ResSAGE builds upon the E-GraphSAGE model by integrating residual connections into the aggregation process, preserving original edge features and mitigating information loss during feature aggregation. This design significantly enhances performance on imbalanced datasets, particularly by improving the representation of minority classes.

E-ResGAT (Chang & Branco, 2024), on the other hand, combines the attention-based mechanisms of GAT with residual learning, further enhancing the representation of both node and edge features. This integration allows E-ResGAT to effectively handle imbalanced data and extract nuanced structural relationships, making it particularly suitable for complex graph-based tasks.

To ensure fair evaluation, the same model architecture and training algorithm were applied across all three datasets: CIC-DarkNet, UNSW-NB15, and ToN-IoT. For the Darknet and UNSW-NB15 datasets, models were trained with consistent hyperparameters. The learning rate was set to 0.0001 for UNSW-NB15, 0.003 for CIC-DarkNet, and 0.0004 for ToN-IoT, and each dataset was divided into training, validation, and test subsets using a 50 %/20 %/30 % split. The Adam optimizer was used for CIC-DarkNet and UNSW-NB15, while the AdamW optimizer was adopted for ToN-IoT to incorporate weight decay regularization. The cross-entropy loss was chosen for its proven efficacy in multi-class classification tasks. To mitigate overfitting, a dropout rate of 0.2 was applied during training for Darknet and UNSW-NB15. For ToN-IoT, higher dropout rates of 0.3 for text embeddings and 0.4 for edge features were employed. These measures provided a robust and reproducible framework for evaluating model performance, particularly in the context of imbalanced datasets.

4.4. Experimental results

The performance of the proposed CG-LLM model was evaluated on the CIC-DarkNet, ToN-IoT, and UNSW-NB15 datasets and compared with two state-of-the-art graph-based models, E-ResSAGE and E-ResGAT (Chang and Branco, 2024), as well as E-GraphSAGE and GAT. To assess the effectiveness of each model, we measured weighted average F1-score, precision, and recall on the multi-classification tasks. The results of this evaluation are summarized in Table 1.

The CIC-DarkNet results demonstrate the superiority of the proposed CG-LLM model across all evaluation metrics. CG-LLM achieves an F1-score of 0.9035, outperforming E-ResGAT (0.8769), E-GraphSAGE (0.8702), GAT (0.8665), and E-ResSAGE (0.8647). In terms of precision, CG-LLM records 0.9368, indicating its ability to minimize false positives, while its recall of 0.8803 reflects strong detection capability across all intrusion categories. These results highlight the effectiveness of integrating contextual node-level embeddings with structural edge features, enabling the model to capture both semantic and topological patterns in darknet traffic. Unlike E-GraphSAGE and E-ResSAGE, which rely solely on edge aggregation, CG-LLM leverages rich language-based embeddings to enhance classification. Although E-ResGAT utilizes attention mechanisms, it lacks the depth of contextual understanding offered by the language model, which proves critical in this complex multi-class task.

In the case of the ToN-IoT dataset, the CG-LLM model also shows competitive performance with an F1-score of **0.9712**, precision of **0.9750**, and recall of **0.9714**. While E-ResGAT (F1-score: 0.9954) and GAT (F1-score: 0.9942) slightly outperform CG-LLM in this dataset, the results still confirm the model's robustness and adaptability to realistic IoT/IloT traffic. The slightly lower performance of CG-LLM on ToN-IoT

Table 1

Comparison of F1-score, Precision, and Recall for multi-classification tasks for CIC-DarkNet, ToN_IoT and UNSW-NB15 datasets across E-GraphSAGE, E-ResSAGE, GAT, E-ResGAT, and CG-LLM.

Dataset	Algorithm	Weighted Average		
		F1-score	Precision	Recall
CIC-DarkNet	E-GraphSAGE	0.8702	0.8782	0.8792
	GAT	0.8665	0.8705	0.8698
	E-ResSAGE	0.8647	0.8723	0.8776
	E-ResGAT	0.8769	0.8874	0.8800
	*CG-LLM	0.9035	0.9368	0.8803
ToN_IoT	E-GraphSAGE	0.9524	0.9555	0.9520
	GAT	0.9942	0.9943	0.9942
	E-ResSAGE	0.9791	0.9799	0.9790
	E-ResGAT	0.9954	0.9955	0.9954
	*CG-LLM	0.9712	0.9750	0.9714
UNSW-NB15	E-GraphSAGE	0.9754	0.9770	0.9808
	GAT	0.9802	0.9792	0.9828
	E-ResSAGE	0.9754	0.9770	0.9808
	E-ResGAT	0.9829	0.9837	0.9844
	*CG-LLM	0.9797	0.9809	0.9802

can be attributed to the nature of the dataset, which may be more structurally regular and less dependent on semantic node contexts. Nonetheless, CG-LLM maintains strong performance without overfitting, showing that the multimodal integration strategy remains effective even in less context-dependent scenarios.

For the **UNSW-NB15** dataset, the CG-LLM model continues to demonstrate robust performance, achieving an F1-score of **0.9797**, precision of **0.9809**, and recall of **0.9802**. This indicates CG-LLM's high effectiveness in accurately identifying attacks while minimizing both false positives and negatives. When compared to other models on UNSW-NB15, such as E-GraphSAGE (F1: 0.9754, Precision: 0.9770, Recall: 0.9808), GAT (F1: 0.9802, Precision: 0.9792, Recall: 0.9828), E-ResSAGE (F1: 0.9754, Precision: 0.9770, Recall: 0.9808), and E-ResGAT (F1: 0.9829, Precision: 0.9837, Recall: 0.9844), CG-LLM remains highly competitive across all evaluation metrics. While E-ResGAT and GAT might show slightly higher F1-scores and recall values in specific instances, CG-LLM's consistent strong performance further validates its multimodal integration strategy. This suggests its strong generalizability across diverse network traffic datasets, including those with varying structural and semantic complexities. The consistently favorable results underscore the benefits of combining contextual language-based embeddings with structural graph features for comprehensive anomaly detection.

The CG-LLM model achieves superior performance across critical classes, as summarized in Table 2, Table 3, and Table 4. The results highlight significant variations across classes, reflecting the challenges posed by multi-label intrusion detection in imbalanced datasets.

As shown in Table 2, the proposed CG-LLM model demonstrates consistent improvements in class-wise F1-score performance on the CIC-DarkNet dataset. In particular, it achieves higher scores in Classes 1, 2, 3, 4, and 5, with F1-scores of 0.8209, 0.6122, 0.8004, 0.5367, and 0.4262, respectively, outperforming state-of-the-art baselines such as E-ResSAGE and E-ResGAT. Notably, CG-LLM substantially outperforms

other models in Classes 1 and 3, which are typically more challenging due to their limited representation and semantic overlap with other classes. These results suggest that the integration of contextual node-level embeddings enables the model to capture subtle semantic distinctions and complex attack behaviors more effectively. While performance on underrepresented classes such as Class 7 and Class 8 remains relatively low, with F1-scores of 0.3207 and 0.1764 respectively, CG-LLM still exceeds competing methods in these categories, indicating enhanced robustness in handling class imbalance.

A comparable trend is observed in the ToN-IoT dataset, as presented in Table 3. Although GAT and E-ResGAT exhibit near-perfect classification on dominant classes, CG-LLM achieves competitive and more balanced performance across a wider range of categories. Specifically, the model yields strong results in Classes 1, 3, and 4, with F1-scores of 0.9476, 0.9537, and 0.8294, respectively, which are known to pose difficulties due to data imbalance or overlapping features with other attack types. By maintaining stable F1-scores across both high- and low-frequency classes, CG-LLM demonstrates its ability to generalize well without overfitting to majority classes. These findings further validate the effectiveness of integrating LLM-derived contextual embeddings with graph-based structural features in complex multi-class intrusion detection tasks.

Like the previous datasets, the UNSW-NB15 dataset also reveals varied per-class F1-score performances, as detailed in Table 4. The proposed CG-LLM model achieves a strong F1-score of 0.9960 for Class 0 (to represent normal traffic), demonstrating excellent detection of benign activities. However, its performance on other classes shows significant variability, recording 0.6114 for Class 1 and 0.6816 for Class 4. For several classes, including 2, 3, 5, 7, 8, and 9, CG-LLM yields an F1-score of 0.0000, indicating a complete inability to detect instances belonging to these categories. This contrasts with other models like GAT, which achieves 0.7368 for Class 1 and 0.8889 for Class 6, and E-ResGAT, which boasts 0.9444 for Class 6 and 0.2500 for Class 8, along with perfect scores of 1.0000 for Class 4. E-ResSAGE also stands out with 0.8461 for Class 2 and a perfect 1.0000 for Class 9. The limitations of CG-LLM on these specific classes highlight challenges in distinguishing certain attack types within the UNSW-NB15 dataset, possibly due to their inherent characteristics, low representation in the dataset, or the model's current inability to capture their unique semantic or structural patterns effectively. Despite its strong overall performance, the per-class results on UNSW-NB15 suggest that further refinement of CG-LLM's contextual understanding or its integration with structural features may be needed to address these specific classification gaps.

To further assess the discriminative capability of the proposed CG-LLM model, we calculated the ROC-AUC scores for each dataset, as summarized in Table 5. The model achieves an overall ROC-AUC of 0.8795 on the CIC-DarkNet dataset, and for the ToN-IoT dataset, the model attains near-perfect performance, with an overall ROC-AUC of 0.9972 and per-class scores consistently exceeding 0.996 across all categories. These results confirm the model's robustness and generalization ability across diverse network intrusion scenarios.

In addition to the overall performance, Fig. 3 presents the ROC curves and per-class ROC-AUC values, illustrating the CG-LLM model's capability to maintain high true positive rates while minimizing false positives. The steep and well-separated curves across classes reflect

Table 2

Comparison of F1-score results per class for CIC-DarkNet dataset across E-GraphSAGE, E-ResSAGE, GAT, E-ResGAT, and CG-LLM.

Model	Per class F1-score								
	0	1	2	3	4	5	6	7	8
E-GraphSAGE	0.9396	0.6563	0.2000	0.5874	0.2807	0.2308	1.0000	0.2029	0.1639
GAT	0.9451	0.6294	0.3200	0.4118	0.3718	0.0000	0.6957	0.0000	0.2800
E-ResSAGE	0.9398	0.6692	0.5556	0.1847	0.3862	0.4117	1.0000	0.3456	0.2727
E-ResGAT	0.9495	0.6373	0.4762	0.4152	0.4167	0.4000	0.8889	0.2903	0.2903
*CG-LLM	0.9389	0.8209	0.6122	0.8004	0.5367	0.4262	0.8095	0.3207	0.1764

Table 3

Comparison of F1-score results per class for ToN-IoT dataset across E-GraphSAGE, E-ResSAGE, GAT, E-ResGAT, and CG-LLM.

Model	Per class F1-score									
	0	1	2	3	4	5	6	7	8	9
E-GraphSAGE	0.9908	0.7281	0.9631	0.7558	0.2727	0.9917	0.8341	0.9355	0.9770	0.8710
GAT	0.9908	0.9769	0.9861	0.9401	1.0000	0.9990	1.0000	1.0000	0.9953	0.9907
E-ResSAGE	0.9907	0.9447	0.9907	0.8525	0.6363	0.9996	0.8801	0.9170	0.9907	0.9723
E-ResGAT	0.9954	0.9816	0.9954	0.9539	1.0000	0.9994	1.0000	1.0000	0.9908	0.9862
*CG-LLM	0.9928	0.9476	0.9582	0.9537	0.8294	0.9980	0.8884	0.8632	0.9050	0.8584

Table 4

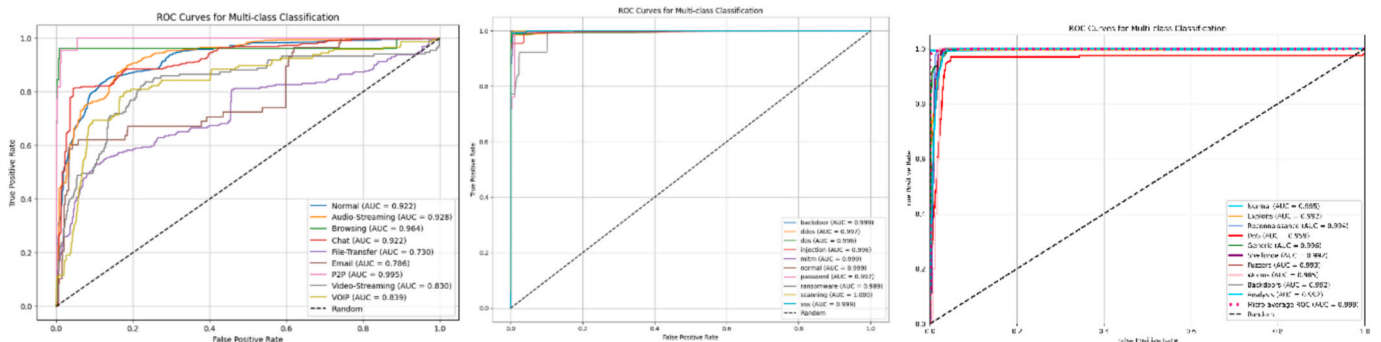
Comparison of F1-score results per class for UNSW-NB15 dataset across E-GraphSAGE, E-ResSAGE, GAT, E-ResGAT, and CG-LLM.

Model	Per class F1-score									
	0	1	2	3	4	5	6	7	8	9
E-GraphSAGE	0.9989	0.6578	0.6923	0.3750	1.0000	0.0000	0.0280	0.0000	0.0000	0.0000
GAT	0.9965	0.7368	0.0000	0.0000	1.0000	0.0000	0.8889	0.0000	0.0000	0.0000
E-ResSAGE	0.9985	0.5000	0.8461	0.0000	1.0000	0.0000	0.3611	0.0000	0.0000	1.0000
E-ResGAT	0.9965	0.6579	0.5385	0.1250	1.0000	0.0000	0.9444	0.0000	0.2500	0.0000
*CG-LLM	0.9960	0.6114	0.0000	0.0000	0.6816	0.0000	0.4527	0.0000	0.0000	0.0000

Table 5

ROC-AUC Performance Comparison in CG-LLM Model across CIC-DarkNet, ToN-IoT, and UNSW-NB15.

Dataset	Per class ROC-AUC										
	Overall	0	1	2	3	4	5	6	7	8	9
CIC-DarkNet	0.8795	0.922	0.928	0.964	0.922	0.730	0.786	0.995	0.830	0.839	—
ToN-IoT	0.9972	0.999	0.997	0.996	0.996	0.996	0.996	0.997	0.999	1.000	0.999
UNSW-NB15	0.9886	0.994	0.993	0.994	0.979	0.995	0.992	0.992	0.981	0.992	0.974

**Fig. 3.** ROC Curve for Multi-Class Classification in CG-LLM Model across CIC-DarkNet, ToN-IoT, and UNSW-NB15.

strong discriminative power, particularly in multi-class settings where class boundaries are not always well-defined. This visualization supports the quantitative results in Table 5 and highlights the effectiveness of combining contextual LLM-based embeddings with structural graph information for enhanced intrusion detection.

Through the integration of node-level contextual embeddings with edge-level structural features, this novel approach provides a scalable and reliable framework for multi-label intrusion detection. Its strong performance across diverse classes underscores its ability to address imbalanced traffic patterns and overcome the limitations of existing graph-based methods like E-ResSAGE and E-ResGAT.

Table 6

Dataset Characteristics and Comparison.

Dataset	Flows ($\times 10^3$)	Classes	Most Frequent Class (%)	Least Frequent Class (%)	Traffic Archetype	Temporal Granularity
CIC-DarkNet	142	9	46.2 (Normal)	0.6 (VOIP)	Mixed Internet, Darknet	Sub-second flow statistics
ToN-IoT	461	10	55.8 (Normal)	0.3 (Ransomware)	IIoT telemetry & enterprise	Millisecond packet traces
UNSW-NB15	700	10	87.3 (Normal)	<0.1 (Worms, Shellcode)	Enterprise perimeter	Second-level summaries

4.5. Robustness and generalizability analysis

4.5.1. Dataset characteristics and Comparison

The CG-LLM model was evaluated across three benchmark datasets: CIC-DarkNet, ToN-IoT, and UNSW-NB15. Each dataset exhibits distinct characteristics that significantly influence model performance (see Table 6). CIC-DarkNet features moderate class imbalance with diverse traffic types, allowing minority classes to remain sufficiently sizeable for effective learning. Endpoint-level features, such as port semantics, provide informative context for the model. In contrast, the ToN-IoT dataset presents more regular and homogeneous IoT/IIoT telemetry, where

structural features dominate, thus diminishing the contextual advantages typically leveraged by CG-LLM. UNSW-NB15 exhibits severe class imbalance, with certain attack classes vastly under-represented and synthetically generated, lacking distinctive endpoint semantics necessary for effective learning.

4.5.2. Performance Variation and underlying Factors

The CG-LLM model demonstrates superior robustness and generalizability under specific scenarios. In conditions of moderate class imbalance with meaningful contextual semantics, such as in the CIC-DarkNet dataset, the model achieves high overall and per-class F1-scores. It consistently outperforms graph neural network (GNN)-only models when distinctions rely heavily on semantic complexity. Additionally, the model excels in scenarios featuring heterogeneous traffic mixtures, where variations in packet size, duration, and directionality enrich contextualization beyond graph structures.

Conversely, CG-LLM encounters significant challenges in scenarios characterized by extreme class sparsity, such as in UNSW-NB15. Classes with less than 0.1 % representation, including Worms and Shellcode, result in negligible model performance ($F1 = 0.0000$). Limited gradient updates, driven by the scarcity of samples and the cross-entropy objective (Buda et al., 2018), cause the model to disproportionately overfit the dominant class (“Normal”). Furthermore, when dealing with semantically homogeneous or synthetically generated traffic, where endpoint identities lose semantic relevance due to anonymization, textual prompts cease to provide informative signals, thereby limiting the contribution of the LLM component. In datasets dominated by structural patterns, such as ToN-IoT, the model remains competitive but does not outperform purely structural GNN architectures (e.g., E-ResGAT, GAT), due to the diminished predictive value of semantic contexts.

4.5.3. Overfitting and Low-Resource class performance

The CG-LLM model consistently struggles with underrepresented classes across all datasets. For example, CIC-DarkNet’s classes 7 and 8 exhibit limited performance gains, with F1-scores as low as 0.1764. In the UNSW-NB15 dataset, several minority classes show complete failure, recording an F1-score of 0.0000. Signs of overfitting are evident, particularly in highly imbalanced contexts such as UNSW-NB15, where precision for minority classes may remain high, yet recall plummets to zero, reflecting overly conservative decision thresholds. Additionally, train-validation curves for minority classes sharply diverge after early epochs.

Although regularization techniques, including dropout and class weighting, have been implemented, they have proven insufficient to fully address extreme class imbalance. Alternative methods such as focal loss (C. Li et al., 2021) and MixUp edge augmentation (Isaksson et al., 2022) showed marginal improvements in minority class performance but negatively impacted macro-F1 scores. Future studies should investigate curriculum sampling or data synthesis methods to better expose models to rare events (Bagui & Li, 2021; Lee et al., 2022). Overall, these findings underline contexts where CG-LLM significantly enhances performance, highlighting ongoing challenges in managing extreme class imbalance, rare events, and evolving threat landscapes in network intrusion detection.

4.6. Interpretability and model explanations via visualization of feature importance

We have employed SHAP-based interpretability (Buda et al., 2018; Lamane et al., 2025; Lundberg & Lee, 2017) to elucidate the contributions of structural features within our graph-based model over the three datasets. We applied SHAP to understand how individual features influence the model’s decisions, thereby contributing to the broader goal of explainable AI by clarifying the reasoning behind each prediction. For the Ton_IoT dataset, our analysis includes global feature importance as well as illustrative example cases, clearly identifying which edge

features influence predictions toward either “malicious” (such as DoS attacks) or “normal.” For the Darknet and UNSW-NB15 datasets, we present one representative case study each, focusing on the Video-Streaming and Generic classes, respectively. This approach balances clarity and conciseness, allowing us to demonstrate the model’s generalizability and interpretability across diverse traffic types without excessive length.

4.6.1. Global SHAP feature importance for DoS attack detection (Ton_IoT dataset)

The bar plot presented ranks the top 15 features contributing significantly to DoS classification, derived from SHAP values (see Fig. 4). This plot reflects class-specific global SHAP importance, computed by averaging SHAP values across all samples labeled as DoS. As a result, individual feature contributions observed in local explanations (e.g., force plots) may differ substantially due to instance-specific conditions. Red bars indicate features increasing the probability of malicious classification, whereas blue bars represent features decreasing this likelihood. Notably, features such as label, missed_bytes, and service significantly drive predictions toward malicious behavior. Additionally, dns_AA, dns_rejected, duration, and bytes also emerge as key contributors.

4.6.2. Example case Study: DoS traffic flow (Ton_IoT dataset)

We analyzed a representative DoS sample using SHAP force plots (see Fig. 5 for SHAP results related to the “DoS” class within the Ton_IoT dataset). The results reveal that specific features, particularly dns_RA, various DNS attributes, and the label significantly contribute towards classifying this flow as “malicious.” Although dns_rejected exhibited a negative SHAP value pushing against a malicious label, the overall classification remained dominated by other stronger indicators. As shown in Fig. 5, the feature DNS-related indicators, such as dns_RA, dns_RD, and dns_AA, contributed positively toward the malicious classification. However, the feature label exhibited a strong negative SHAP value (−2.54), significantly lowering the prediction, likely due to its association with benign traffic in this instance. Similarly, dns_rejected contributed negatively (−0.48). These opposing signals moderated the final prediction, resulting in a moderate malicious score of approximately 0.46. These findings imply that malicious flows such as DoS attacks demonstrate distinctive DNS behaviors, substantial traffic volumes (bytes, missed flows), and particular connection metadata patterns. Awareness of these features allows incident responders to prioritize attention on packets exhibiting similar anomalies, thereby streamlining triage and forensic efforts.

4.6.3. Example case Study: Normal traffic flow (Ton_IoT dataset)

An analysis of a typical “normal” traffic flow, alongside its associated

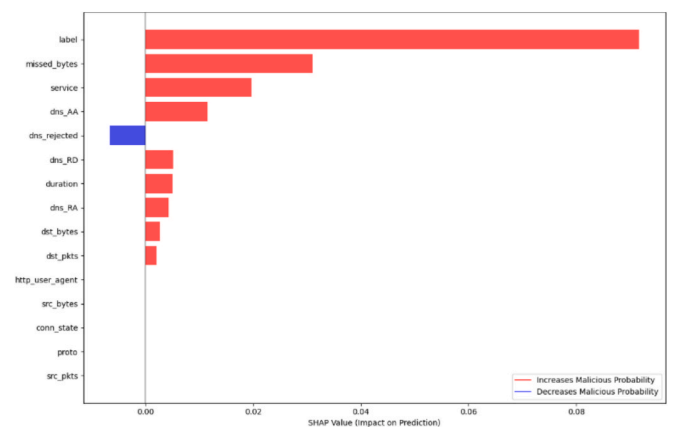


Fig. 4. Top 15 features contributing to DoS classification based on class-conditional global SHAP analysis.

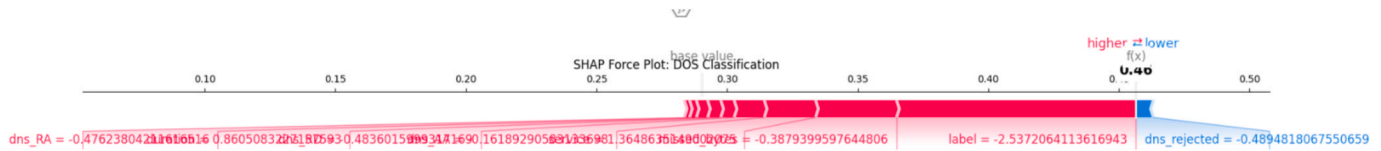


Fig. 5. SHAP force plot for DoS class within Ton_IoT dataset.

feature importances, was conducted (see Fig. 6 for SHAP results corresponding to the “Normal” class within the Ton_IoT dataset). The top ten consistent features contributing to normal classification include *dst_bytes*, *timestamp* (*ts*), *service*, *duration*, *conn_state*, *missed_bytes*, *src_ip_bytes*, *src_pkts*, *src_bytes*, and *protocol* (*proto*) (see Fig. 7). Among these, *dst_bytes* displays a notably negative SHAP value, significantly reducing the likelihood of malicious classification (approximately -0.0932). In contrast, features such as *timestamp* (*ts*), *service*, and *duration* contribute positively, with SHAP values around 0.0382, 0.0313, and 0.0184 respectively, thereby supporting the classification as normal. Anomalies related to DNS and connection attributes do not significantly impact normal predictions, reaffirming that the model prioritizes traffic volume and protocol context to distinguish benign activities. For practitioners, these observations emphasize the necessity of establishing properly profiled normal traffic baselines. Significant deviations in these contextual features may indicate potential incidents and warrant further investigation.

It is important to note that while the bar chart in Fig. 7 reflects average feature contributions across all samples labeled as normal, the force plot in Fig. 6 presents the SHAP values for a single instance. As a result, the most influential features observed locally may not fully align with global patterns due to instance-specific conditions. In this case, the model’s low prediction score (approximately 0.01) is driven by several features exerting strong negative SHAP values, while only one feature has a small opposing influence. This further supports the global trend that benign traffic is typically associated with volume- and protocol-level characteristics.

Local explanation analysis on the Ton_IoT dataset reveals clear distinctions between malicious and normal traffic. Features such as unusual DNS flags and high missed byte counts push predictions toward malicious classifications, while expected values for protocol and volume-related features reduce the likelihood of false positives. These explanations align with real-world cybersecurity workflows, supporting the interpretability of model decisions in incident response, threshold tuning, and compliance contexts (Adhikari & Thapaliya, 2024; Guerra-Manzanares et al., 2019; Z. Zhang et al., 2022), while also enhancing the transparency and auditability of machine learning-based detection systems.

4.6.4. Example case Study: Video-streaming traffic flow (Darknet Dataset)

To further understand the model’s decision-making process, we conducted a SHAP analysis on a representative sample predicted as Video-Streaming from the CIC-Darknet dataset. As shown in Fig. 8, the feature “Protocol” exhibited the strongest positive SHAP impact (approximately $+0.78$), significantly increasing the model’s confidence in the video-streaming classification. Additional contributing features include “Bwd Packet Length Min,” “FIN Flag Count,” and “Fwd Packet

Length Std,” reflecting temporal and flow-level characteristics commonly associated with streaming activity. The instance-level prediction score was approximately 0.50, indicating a moderately confident classification, influenced by the cumulative effect of multiple smaller features. Fig. 9 summarizes the top ten most consistent features for the class based on average SHAP values across all samples. These findings illustrate the model’s ability to capture streaming-related flow behaviors in a mixed traffic environment, using both protocol context and statistical packet descriptors.

4.6.5. Example case Study: Generic traffic flow (UNSW-NB15 Dataset)

The SHAP case study for UNSW-NB15 dataset reveals that *ct_state_ttl* (connection state time-to-live) emerges as the most dominant feature for Generic traffic classification with a mean SHAP value of 0.1452, significantly outweighing all other features (see Fig. 11). This suggests that connection state characteristics and TTL behaviors are highly distinctive markers for generic network activities. The second most influential feature, *sttl* (source TTL) with a SHAP value of 0.0685, indicates that time-to-live values from the source are crucial for distinguishing generic traffic patterns. Other notable contributors include *dwin* (destination window size, 0.0474) and *dttl* (destination TTL, 0.0438), highlighting the importance of both source and destination characteristics in classification decisions. The presence of *dmeansz* (destination mean size, 0.0433) and *ltime* (last time, 0.0386) further emphasizes that packet size patterns and temporal features play significant roles. The inclusion of packet-level features like *spkts* (source packets) and *sloss* (source loss) indicates that the model effectively captures both connection characteristics and transmission quality metrics to distinguish generic network behavior from other traffic classes. As shown in Fig. 10, the SHAP force plot for a representative instance further illustrates this decision by highlighting *ct_state_ttl* as the dominant feature, alongside strong contributions from *spkts*, *dmeansz*, and *sloss*, cumulatively pushing the model output toward the generic class with high confidence. The force plot complements the global ranking in Fig. 11 by revealing how these features interact in an individual instance to produce a high-confidence classification.

4.6.6. Implications across datasets

The SHAP-based interpretability results presented across the Ton_IoT, CIC-Darknet, and UNSW-NB15 datasets demonstrate that our graph-based model yields explainable and consistent outputs in varied network contexts. These results affirm that the model’s decisions are driven by well-defined edge features, supporting both transparency and operational usability. Such interpretability is essential in cybersecurity settings where automated decisions must be accountable, traceable, and aligned with domain expectations.

Across all datasets, the model leverages a mix of structural features,

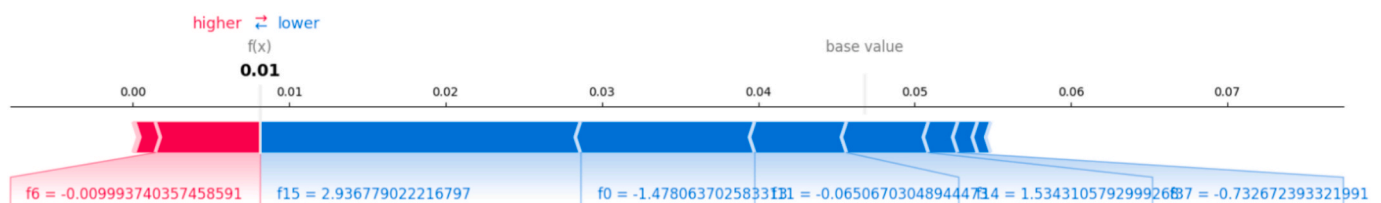


Fig. 6. SHAP force plot for normal class within Ton_IoT dataset.

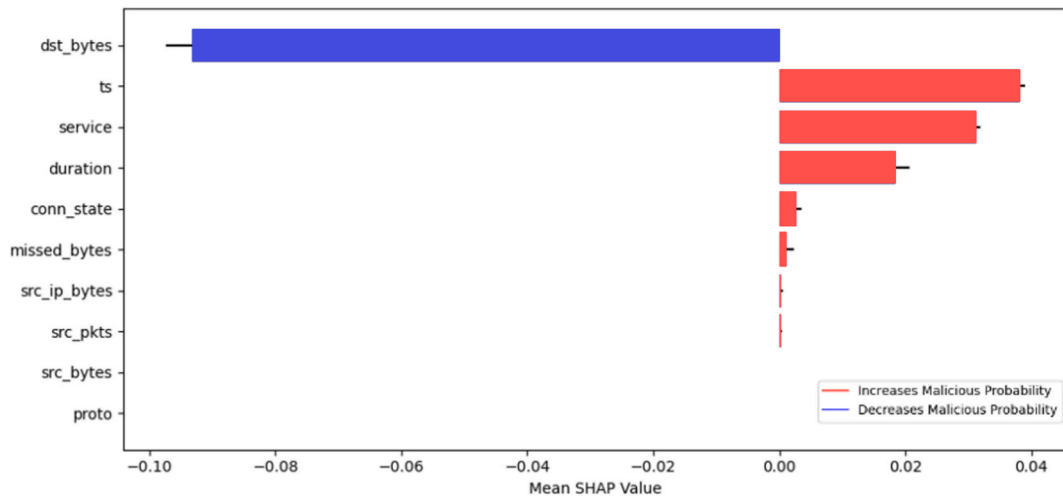


Fig. 7. Top 10 important features for normal class within TonIoT dataset.

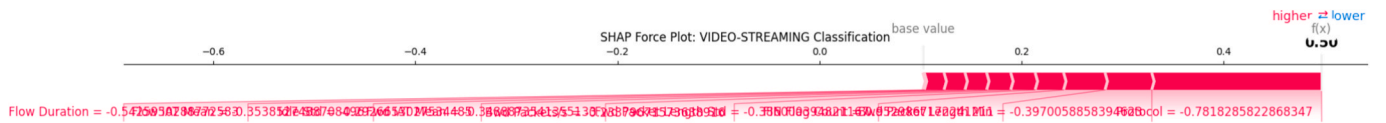


Fig. 8. SHAP force plot for video-streaming class within Darknet dataset.

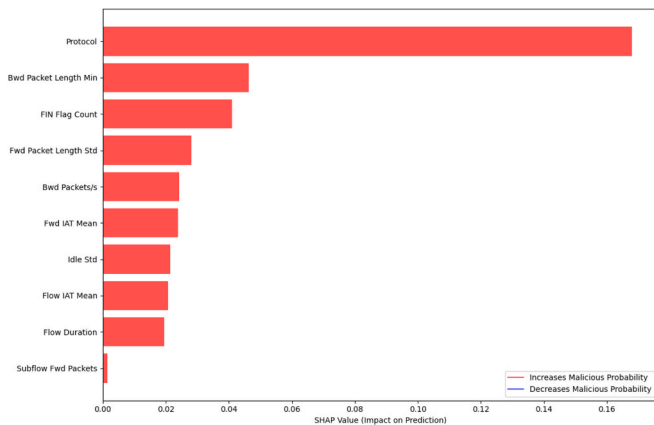


Fig. 9. Top 10 important features for video-streaming class within Darknet dataset.

such as traffic volume indicators, connection-state attributes, and timing-related measures, to inform its predictions. While the exact set of dominant features varies by dataset, the presence of flow-level and contextual indicators in each case points to a shared decision-making pattern. This suggests a degree of generalizability in how the model captures behavioral traits of network traffic. Rather than relying on isolated or dataset-specific cues, the model identifies recurring structural signals, such as byte counts, duration, or TTL-related attributes, that contribute meaningfully across settings.

At the same time, each dataset also revealed context-specific decision profiles. For TonIoT, DNS-related attributes and missed byte counts

dominated DoS classifications. In the CIC-Darknet dataset, protocol usage and packet length variability were most indicative of video-streaming traffic. In UNSW-NB15, connection lifetime and TTL-based features shaped the model's understanding of Generic flows. These findings illustrate that while the model learns general traffic representations, it also captures specialized behaviors relevant to the dataset's operational setting. This combination of transferable core logic and adaptive detail reinforces the model's practical applicability in diverse network environments.

The implications of these findings are significant for deployment and trust in machine learning-based detection systems. SHAP-based explanations allow analysts to identify key indicators behind each prediction, improving the speed and confidence of triage and investigation. The interpretability can further support compliance and governance requirements by clarifying the basis for automated decisions. Moreover, the observed generalization across datasets provides evidence that the model is not overfitted to a single traffic profile, but rather capable of adapting to unseen contexts by preserving relevant structural features.

In summary, the integration of global and local interpretability via SHAP not only enhances the explainability of model behavior but also substantiates its generalizability. This dual capability strengthens the foundation for deploying machine learning in dynamic and heterogeneous cybersecurity environments, enabling more informed, robust, and justifiable decision-making.

4.7. Computational complexity analysis

The proposed framework integrates transformer-based LLMs and GNNs to effectively capture both semantic relationships and structural dependencies within cybersecurity graphs. The computational

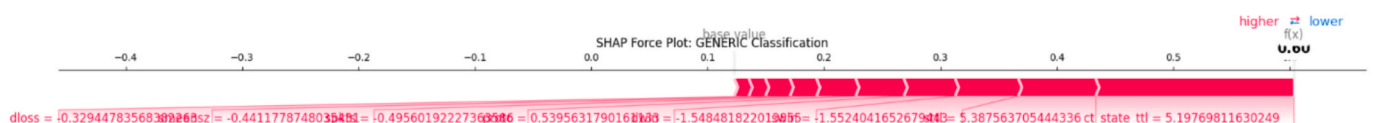


Fig. 10. SHAP force plot for generic class within UNSW-NB15 dataset.

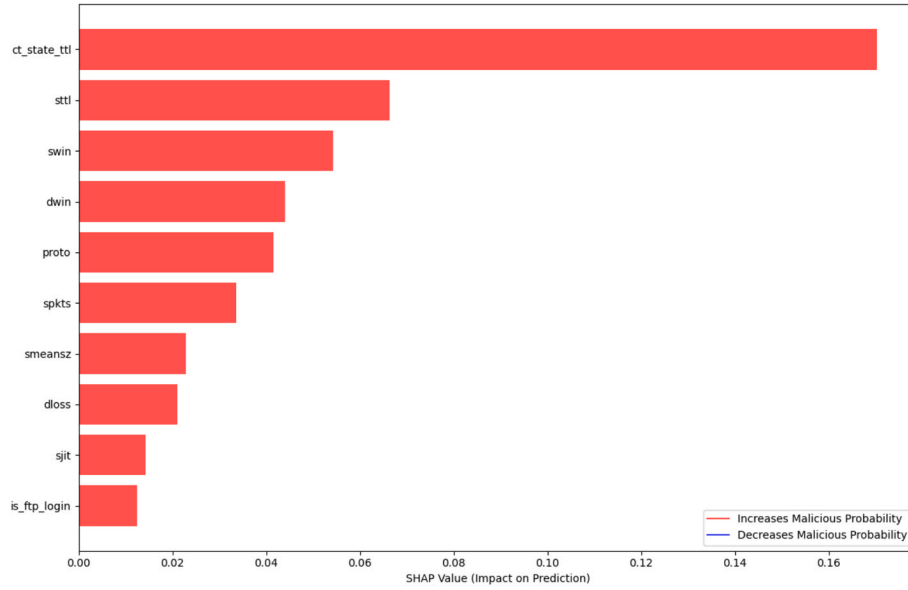


Fig. 11. Top 10 important features for generic class within UNSW-NB15 dataset.

complexity of the framework can be decomposed into the following components.

- (1) **Node Embedding via LLM:** Given N nodes, the model constructs textual queries for node pairs and processes them using a pre-trained transformer-based LLM. For a hidden dimension d and maximum sequence length L , the complexity of a single query is approximately $O(L^2 \cdot d)$. Assuming pairwise inference over PPP node pairs, the total LLM-related complexity becomes $O(P \cdot L^2 \cdot d)$. In practice, P is significantly reduced through edge sampling or sparsification techniques.
- (2) **GNN-based Edge Feature Aggregation:** For a graph with NNN nodes and EEE edges, the GNN performs neighborhood aggregation using edge features. The computational complexity per GNN layer is $O(E \cdot d)$, where ddd denotes the embedding dimension.
- (3) **Classifier Layer:** The final feedforward classification layer incurs negligible cost with complexity $O(N \cdot d)$.
- (4) **Regularization Techniques:** Dropout (with rates between 0.2 and 0.4) is applied during training to improve generalization. These techniques marginally increase training time but do not impact the asymptotic complexity.
- (5) **Total Complexity:** The overall computational complexity of the framework can be expressed as:

$$O(P \cdot L^2 \cdot d + E \cdot d) \quad (7)$$

This is higher than conventional GNN models due to the LLM component; however, it remains computationally feasible for real-world datasets, especially with the use of precomputed embeddings and mini-batch processing.

To demonstrate the practicality of the proposed method, training on the CIC-DarkNet (141,530 samples), ToN-IoT (461,043 samples), and UNSW-NB15 (700,001 samples) datasets was conducted on an NVIDIA H100 HBM3 GPU with 80 GB of memory. In Table 7, for the ToN-IoT dataset, training was completed in approximately 29 min and 50 s, with peak memory usage reaching approximately 25.82 GB. For the CIC-DarkNet dataset, training required approximately 2 h and 18 min, with peak GPU memory usage of 23.77 GB. Similarly, the UNSW-NB15 dataset completed training in about 2 h and 18 min, with a peak usage of 27.00 GB. These measurements highlight the scalability and computational efficiency of the CG-LLM model, even when applied to

Table 7

Training Time and Peak GPU Memory Usage of CG-LLM across Datasets.

Dataset	Number of Samples	Training Time (hh:mm:ss)	Peak Memory Usage (GB)
CIC-DarkNet	141,530	02:18:49	23.77
ToN-IoT	461,043	00:29:50	25.82
UNSW-NB15	700,001	02:18:20	27.00

large-scale and complex datasets.

4.8. Ablation study

An ablation study systematically evaluates individual components of a model to understand their contributions to overall performance (Meyes et al., 2019; Sheikholeslami et al., 2021). We conducted an ablation study to assess the effectiveness and generalizability of the proposed CG-LLM model by comparing it to a standard GNN baseline. We did not include an “LLM-only” variant because our approach specifically integrates contextual features derived from LLM embeddings with numerical and structural features representing network traffic statistics. Without protocol and flow-level numerical data, an LLM-only model would lack sufficient discriminatory power, given that malicious behaviors often manifest as statistical anomalies rather than purely semantic or textual cues.

The comparative results across three benchmark datasets, CIC-Darknet, ToN-IoT, and UNSW-NB15, are summarized in Table 8. CG-LLM consistently demonstrates improved or competitive performance compared to the baseline GNN model, although the extent of improvement varies depending on dataset characteristics and class distributions.

On the CIC-Darknet dataset, CG-LLM significantly outperforms the GNN baseline in terms of weighted F1-score (0.9035 vs. 0.8623), precision, and recall. This improvement is further highlighted in the per-class results (Table 9), where CG-LLM achieves notable gains on several challenging and minority classes, thereby enhancing the balance and robustness of intrusion detection.

For the ToN-IoT dataset, both models achieve high accuracy, yet CG-LLM attains slightly higher weighted averages for F1-score (0.9712 vs. 0.9700), precision, and recall. Importantly, CG-LLM reduces the performance gap for difficult-to-detect classes, demonstrating better

Table 8

Comparison of F1-score, Precision and Recall results for three datasets across, GNN only and CG-LLM.

Dataset	Model	Weighted Average		
		F1-score	Precision	Recall
CIC-DarkNet	GNN only	0.8623	0.8637	0.8690
	*CG-LLM	0.9035	0.9368	0.8803
UNSW-NB15	GNN only	0.9865	0.9892	0.9858
	*CG-LLM	0.9797	0.9809	0.9802
ToN_IoT	GNN only	0.9700	0.9721	0.9698
	*CG-LLM	0.9712	0.9750	0.9714

generalization capabilities across diverse traffic types.

On the UNSW-NB15 dataset, however, the baseline GNN model shows marginally better performance in weighted F1-score (0.9865 vs. 0.9797), particularly for major classes. Both models struggle significantly with minority or complex classes (e.g., classes 5–9), exhibiting near-zero F1-scores. These results suggest that CG-LLM provides limited additional benefit in contexts where conventional GNNs already perform effectively.

Overall, these findings indicate that CG-LLM offers substantial advantages for datasets characterized by class imbalance and diverse traffic (CIC-Darknet and ToN-IoT), especially in improving minority-class detection. While the results on UNSW-NB15 highlight scenarios where improvements are less pronounced, they do not undermine the value of CG-LLM. Instead, they clarify the contexts in which integrating semantic and structural features is most beneficial, thus affirming the practical applicability of CG-LLM in realistic cybersecurity environments.

4.9. Discussion, Limitations, and future work

The proposed CG-LLM model introduces a novel framework that integrates structural graph analysis with textual contextualization using TinyLlama for intrusion detection. Unlike existing methods, which rely solely on structural graph features, this approach combines node-level textual embeddings and graph-based representations to capture complex, multi-dimensional relationships in network data. This fusion achieves superior classification performance, particularly on imbalanced datasets like CIC-DarkNet, ToN-IoT, and UNSW-NB15, with notable improvements in detecting underrepresented attack types.

The study's innovative methodology lies in embedding textual insights from network nodes into the graph structure, facilitated by the compact and efficient TinyLlama model. This enables contextual understanding of node attributes, such as behaviors or packet flows from various source IP addresses, enriching the graph representation. By transforming network flow data into a hybrid structure combining graph topology and its context, the model demonstrates robust intrusion detection capabilities, outperforming prior state-of-the-art approaches.

This study underscores the value of combining structural and contextual methods in network intrusion detection, setting the stage for more sophisticated multimodal solutions in cybersecurity. The main implications of this research that highlights the potential of integrating LLMs into graph-based cybersecurity solutions are offering better

scalability and interpretability. The CG-LLM approach addresses challenges like data imbalance and high dimensionality while providing a scalable solution suitable for real-world applications. By leveraging open-source tools and datasets, this work fosters reproducibility and serves as a foundation for further innovation in Darknet traffic analysis.

One limitation of the current approach is the absence of explicit modeling of temporal dependencies or flow sequences. While the CG-LLM architecture leverages static graph structures, it incorporates time-based flow statistics through edge features from all three evaluated datasets. These include direct timing features, and indirect indicators of temporal behavior. While these features are used in aggregate form and do not constitute temporal modeling in a strict sequential sense, they contribute to capturing latent time-based behavior in edge-level encoding. Furthermore, the integration of a LLM introduces a computational overhead, posing challenges for resource-constrained environments. Computational demands associated with LLM inference and training, as evidenced by the peak GPU memory usage ranging from 23.77 GB to 27.00 GB and training times up to 2 h and 18 min need to be carefully considered for practical deployment. Despite its overall strong performance, the model can still face challenges in classifying extremely rare attack types, where the scarcity of data limits the learning capability. Extending the framework with dynamic temporal models or sequence encoders remains the valuable direction for feature work.

It is important to emphasize that CG-LLM is currently a research-stage framework and not yet optimized for immediate real-world deployment. Future directions for this research include integrating temporal graph learning to explicitly model dynamic relationships and flow sequences, and developing efficient fine-tuning strategies for LLMs specifically tailored for cybersecurity applications. Additionally, exploring lightweight variants of the CG-LLM architecture for edge deployment and investigating domain adaptation techniques to generalize the model to new network environments are crucial next steps.

5. Conclusion

This paper introduced the CG-LLM framework, which combines graph-based structural representations with LLM-driven textual embeddings to address the increasing complexity of Darknet network traffic. By transforming raw flow records from the CIC-DarkNet, ToN-IoT, and UNSW-NB15 into a bipartite graph and pairing them with node-level contextual information derived from TinyLlama, CG-LLM offers a richer, more nuanced understanding of network activities. Our experiments demonstrate that CG-LLM significantly outperforms leading graph-based methods—including E-GraphSAGE, GAT, E-ResSAGE, and E-ResGAT—especially in handling underrepresented attack classes. These findings underscore the value of fusing structural and semantic insights, revealing that multimodal approaches can better detect subtle anomalies and enhance multi-label intrusion detection performance. Nevertheless, incorporating an LLM poses computational challenges, prompting future research on optimizing training and inference for large-scale or resource-constrained environments. Moreover, while the present work centers on a single dataset, broader validation against diverse Darknet datasets and real-world deployments would reinforce its generalizability. Overall, the successful application of CG-LLM

Table 9

Comparison of F1-score results per class for three datasets across, GNN only and CG-LLM.

Dataset	Model	Per class F1-score									
		0	1	2	3	4	5	6	7	8	9
Darknet	GNN only	0.9478	0.5940	0.5111	0.0040	0.2620	0.0000	0.6923	0.2725	0.1608	
	*CG-LLM	0.9389	0.8209	0.6122	0.8004	0.5367	0.4262	0.8095	0.3207	0.1764	
UNSW-NB15	GNN only	0.9968	0.6678	0.6600	0.0169	0.9563	0.0000	0.5750	0.0000	0.0000	
	*CG-LLM	0.9960	0.6114	0.0000	0.0000	0.6816	0.0000	0.4527	0.0000	0.0000	0.5723
ToN-IoT	GNN only	0.9980	0.9196	0.9208	0.9715	0.6937	0.9981	0.8056	0.7935	0.9950	0.9476
	*CG-LLM	0.9928	0.9476	0.9582	0.9537	0.8294	0.9980	0.8884	0.8632	0.9050	0.8584

highlights the potential of integrating GNNs with LLMs to bolster cybersecurity measures in anonymized, high-stakes environments, marking a step forward in proactive threat detection and network defense.

6. Declaration

We corrected the grammatical errors and rephrase some sentences by using large language model (ChatGPT4o).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We used publicly available datasets

References

- Abloh, L., & Libicki, M. (2015). Hacker's Bazaar: The Markets for Cybercrime Tools and Stolen Data. *Defense Counsel Journal*, 82, 143.
- Abu Al-Haija, Q., Krichen, M., & Abu Elhaija, W. (2022). Machine-Learning-Based Darknet Traffic Detection System for IoT Applications. *Electronics*, 11(4), Article 4. <https://doi.org/10.3390/electronics11040556>
- Adhikari, D., & Thapaliya, S. (2024). Explainable AI for Cyber Security: Interpretable Models for Malware Analysis and Network Intrusion Detection. *NPRC Journal of Multidisciplinary Research*, 1(9), Article 9. <https://doi.org/10.3126/nprcjmrvli9.74177>
- Adjewa, F., Essegir, M., & Merghem-Boulahia, L. (2024). LLM-based Continuous Intrusion Detection Framework for Next-Gen Networks (arXiv:2411.03354). arXiv. doi: 10.48550/arXiv.2411.03354.
- Alimoradi, M., Zabihimayvan, M., Daliri, A., Sledzik, R., & Sadeghi, R. (2022). Deep Neural Classification of Darknet Traffic. In *Artificial Intelligence Research and Development* (pp. 105–114). IOS Press. <https://doi.org/10.3233/FAIA220323>.
- Almomani, A. (2023). Darknet traffic analysis, and classification system based on modified stacking ensemble learning algorithms. *Information Systems and E-Business Management*. <https://doi.org/10.1007/s10257-023-00626-2>
- Alsaedi, A., Moustafa, N., Tari, Z., Mahmood, A., & Anwar, A. (2020). TON IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems. *IEEE Access*, 8, 165130–165150. <https://doi.org/10.1109/ACCESS.2020.3022862>
- Ashraf, J., Keshk, M., Moustafa, N., Abdel-Basset, M., Khurshid, H., Bakhshi, A. D., & Mostafa, R. R. (2021). IoTBot-IDS: A novel statistical learning-enabled botnet detection framework for protecting networks of smart cities. *Sustainable Cities and Society*, 72, Article 103041. <https://doi.org/10.1016/j.scs.2021.103041>
- Bagui, S., & Li, K. (2021). Resampling imbalanced data for network intrusion detection datasets. *Journal of Big Data*, 8(1), 6. <https://doi.org/10.1186/s40537-020-00390-x>
- Bilot, T., Madhoun, N. E., Agha, K. A., & Zouaoui, A. (2023). Graph Neural Networks for Intrusion Detection: A Survey. *IEEE Access*, 11, 49114–49139. *IEEE Access*. doi: 10.1109/ACCESS.2023.3275789.
- Booi, T. M., Chiscop, I., Meeuwissen, E., Moustafa, N., & den Hartog, F. T. H. (2022). ToN IoT: The Role of Heterogeneity and the Need for Standardization of Features and Attack Types in IoT Network Intrusion Data Sets. *IEEE Internet of Things Journal*, 9(1), 485–496. <https://doi.org/10.1109/IIOT.2021.3085194>
- Bradley, A. P. (1997). The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognition*, 30(7), 1145–1159. [https://doi.org/10.1016/S0031-3203\(96\)00142-2](https://doi.org/10.1016/S0031-3203(96)00142-2)
- Buda, M., Maki, A., & Mazurowski, M. A. (2018). A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks*, 106, 249–259. <https://doi.org/10.1016/j.neunet.2018.07.011>
- Chang, L., & Branco, P. (2024). Embedding residuals in graph-based solutions: The E-ResSAGE and E-ResGAT algorithms. A case study in intrusion detection. *Applied Intelligence*, 54(8), 6025–6040. <https://doi.org/10.1007/s10489-024-05404-2>
- Chen, Y., Li, R., Zhao, Z., & Zhang, H. (2024). MERLOT: A Distilled LLM-based Mixture-of-Experts Framework for Scalable Encrypted Traffic Classification, arXiv:2411.13004, arXiv. <https://doi.org/10.48550/arXiv.2411.13004>
- Christin, N. (2013). Traveling the silk road: A measurement analysis of a large anonymous online marketplace. In *Proceedings of the 22nd International Conference on World Wide Web* (pp. 213–224). <https://doi.org/10.1145/2488388.2488408>
- Curebal, F., & Dag, H. (2024). Enhancing Malware Classification: A Comparative Study of Feature Selection Models with Parameter Optimization. *Systems and Information Engineering Design Symposium (SIEDS)*, 2024, 511–516. <https://doi.org/10.1109/SIEDS61124.2024.10534669>
- Cürebal, F., Demirkıran, F., Çayır, A., & Dağ, H. (2023). Correlation Matrix as a Smart Filter for Malware Classification Using Ensemble of Novel Feature Selection Algorithms. *Research Square*. <https://doi.org/10.21203/rs.3.rs-3154479/v1>
- Demertzis, K., Tsiknas, K., Takezis, D., Skianis, C., & Iliadis, L. (2021). Darknet Traffic Big-Data Analysis and Network Management for Real-Time Automating of the Malicious Intent Detection Process by a Weight Agnostic Neural Networks Framework. *Electronics*, 10(7), Article 7. <https://doi.org/10.3390/electronics10070781>
- Deng, X., Zhu, J., Pei, X., Zhang, L., Ling, Z., & Xue, K. (2023). Flow Topology-Based Graph Convolutional Network for Intrusion Detection in Label-Limited IoT Networks. *IEEE Transactions on Network and Service Management*, 20(1), 684–696. *IEEE Transactions on Network and Service Management*. doi: 10.1109/TNSM.2022.3213807.
- Duan, G., Lv, H., Wang, H., Feng, G., & Li, X. (2024). Practical Cyber Attack Detection With Continuous Temporal Graph in Dynamic Network System. *IEEE Transactions on Information Forensics and Security*, 19, 4851–4864. *IEEE Transactions on Information Forensics and Security*. doi: 10.1109/TIFS.2024.3385321.
- Farrukh, Y. A., Wali, S., Khan, I., & Bastian, N. D. (2024). XG-NID: Dual-Modality Network Intrusion Detection using a Heterogeneous Graph Neural Network and Large Language Model, arXiv:2408.16021, arXiv. <https://doi.org/10.48550/arXiv.2408.16021>
- Fatani, A., Abd Elaziz, M., Dahou, A., Al-Qaness, M. A. A., & Lu, S. (2021). IoT Intrusion Detection System Using Deep Learning and Enhanced Transient Search Optimization. *IEEE Access*, 9, 123448–123464. *IEEE Access*. doi: 10.1109/ACCESS.2021.3109081.
- Ganapathyappan, K., & Yogish, A. (2024). Detection and Characterization of Darknet Traffic Using Attention LSTM with XAI. *International Conference on Intelligent Systems for Cybersecurity (ISCS)*, 2024, 1–7. <https://doi.org/10.1109/ISCS61804.2024.10581210>
- Guerra-Manzanares, A., Nömm, S., & Bahsi, H. (2019). Towards the Integration of a Post-Hoc Interpretation Step into the Machine Learning Workflow for IoT Botnet Detection. In *2019 18th IEEE International Conference On Machine Learning And Applications (ICMLA)* (pp. 1162–1169). <https://doi.org/10.1109/ICMLA.2019.00193>
- Habibi Lashkari, A., Kaur, G., & Rahali, A. (2020). DIDarknet: A Contemporary Approach to Detect and Characterize the Darknet Traffic using Deep Image Learning. In *2020 the 10th International Conference on Communication and Network Security* (pp. 1–13). <https://doi.org/10.1145/3442520.3442521>
- Hand, D. J., & Till, R. J. (2001). A Simple Generalisation of the Area Under the ROC Curve for Multiple Class Classification Problems. *Machine Learning*, 45(2), 171–186. <https://doi.org/10.1023/A:1010920819831>
- Hu, X., Li, N., Cheng, G., Li, R., & Wu, H. (2023). A Novel Darknet Traffic Classification Method Based on Knowledge Graph with Dynamic Embedding Learning. In *ICC 2023 - IEEE International Conference on Communications* (pp. 3799–3804). <https://doi.org/10.1109/ICC45041.2023.10279209>
- Hwang, R.-H., Peng, M.-C., Huang, C.-W., Lin, P.-C., & Nguyen, V.-L. (2020). An Unsupervised Deep Learning Model for Early Network Traffic Anomaly Detection. *IEEE Access*, 8, 30387–30399. *IEEE Access*. doi: 10.1109/ACCESS.2020.2973023.
- IBM. (2024). IBM Security X-Force Threat Intelligence Index 2024. <https://www.ibm.com/reports/threat-intelligence>.
- Isaksson, L. J., Summers, P., Raimondi, S., Gandini, S., Bhalerao, A., Marvaso, G., Petralia, G., Pega, M., & Jereczek-Fossa, B. A. (2022). Mixup (Sample Pairing) Can Improve the Performance of Deep Segmentation Networks. *Journal of Artificial Intelligence and Soft Computing Research*, 12(1), 29–39. <https://doi.org/10.2478/jaiscr-2022-0003>
- Kevric, J., Jukic, S., & Subasi, A. (2017). An effective combining classifier approach using tree algorithms for network intrusion detection. *Neural Computing and Applications*, 28(1), 1051–1058. <https://doi.org/10.1007/s00521-016-2418-1>
- Lamane, H., Mouhir, L., Moussadek, R., Baghdad, B., Kisi, O., & El Bilali, A. (2025). Interpreting machine learning models based on SHAP values in predicting suspended sediment concentration. *International Journal of Sediment Research*, 40(1), 91–107. <https://doi.org/10.1016/j.ijsrc.2024.10.002>
- Lan, J., Liu, X., Li, B., Li, Y., & Geng, T. (2022). DarknetSec: A novel self-attentive deep learning method for darknet traffic classification and application identification. *Computers & Security*, 116, Article 102663. <https://doi.org/10.1016/j.cose.2022.102663>
- Lashkari, A. H., Kaur, G., & Rahali, A. (2021). DIDarknet: A Contemporary Approach to Detect and Characterize the Darknet Traffic using Deep Image Learning. In *Proceedings of the 2020 10th International Conference on Communication and Network Security* (pp. 1–13). <https://doi.org/10.1145/3442520.3442521>
- Lee, B.-S., Kim, J.-W., & Choi, M.-J. (2022). Experimental Comparison of Hybrid Sampling Methods for an Efficient NIDS. In *2022 23rd Asia-Pacific Network Operations and Management Symposium (APNOMS)* (pp. 1–4). <https://doi.org/10.23919/APNOMS6106.2022.9919939>
- Li, C., Chen, M., Zhang, J., & Liu, H. (2021). Cardiac MRI segmentation with focal loss constrained deep residual networks. *Physics in Medicine & Biology*, 66(13), Article 135012. <https://doi.org/10.1088/1361-6560/ac0bd3>
- Li, H., & Chasaki, D. (2024). Heterogeneous GNN with Express Edges for Intrusion Detection in Cyber-Physical Systems. In *2024 International Conference on Computing, Networking and Communications (ICNC)* (pp. 523–529). <https://doi.org/10.1109/ICNC59896.2024.10556029>
- Li, J., & Pan, Z. (2020). Network Traffic Classification Based on Deep Learning. *KSII Transactions on Internet and Information Systems (TIIS)*, 14(11), 4246–4267. <https://doi.org/10.3837/tiis.2020.11.001>
- Li, P., Zhao, W., Wang, C., Xia, Q., Wu, A., & Peng, L. (2023). Practice with Graph-based ANN Algorithms on Sparse Data: Chi-square Two-tower model, HNSW, Sign Cauchy. *Projections*, arXiv:2306.07607, arXiv. <https://doi.org/10.48550/arXiv.2306.07607>
- Li, Y., Xiang, Z., Bastian, N. D., Song, D., & Li, B. (2024, October 22). IDS-Agent: An LLM Agent for Explainable Intrusion Detection in IoT Networks. *NeurIPS 2024 Workshop on Open-World Agents*. <https://openreview.net/forum?id=iIK0PrYlkW>.

- Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., & Neubig, G. (2023). Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing. *ACM Comput. Surv.*, 55(9), 195: 1–195, 35. <https://doi.org/10.1145/3560815>
- Liu, S., Yao, D., Fang, L., Li, Z., Li, W., Feng, K., Ji, X., & Bi, J. (2024). *AnomalyLLM: Few-shot Anomaly Edge Detection for Dynamic Graphs using Large Language Models*, arXiv: 2405.07626, arXiv. <https://doi.org/10.48550/arXiv.2405.07626>
- Lo, W. W., Layeghy, S., Sarhan, M., Gallagher, M., & Portmann, M. (2022). E-GraphSAGE: A Graph Neural Network based Intrusion Detection System for IoT. In *NOMS 2022-2022 IEEE/IFIP Network Operations and Management Symposium* (pp. 1–9). <https://doi.org/10.1109/NOMS54207.2022.9789878>
- Lücke, J., & Eggert, J. (2010). Expectation truncation and the benefits of preselection in training generative models. *The Journal of Machine Learning Research*, 11, 2855–2900.
- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*, 30. <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- Mahboubi, A., Luong, K., Aboutorab, H., Bui, H. T., Jarrad, G., Bahutair, M., Camtepe, S., Pogrebna, G., Ahmed, E., Barry, B., & Gately, H. (2024). Evolving techniques in cyber threat hunting: A systematic review. *Journal of Network and Computer Applications*, 232, Article 104004. <https://doi.org/10.1016/j.jnca.2024.104004>
- Meidan, Y., Bohadana, M., Shabtai, A., Guarnizo, J. D., Ochoa, M., Tippenhauer, N. O., & Elovici, Y. (2017). ProfilloT: A machine learning approach for IoT device identification based on network traffic analysis. In *Proceedings of the Symposium on Applied Computing* (pp. 506–509). <https://doi.org/10.1145/3019612.3019878>
- Meyes, R., Lu, M., Puisseau, C. W. de, & Meisen, T. (2019). Ablation Studies in Artificial Neural Networks (arXiv:1901.08644). arXiv. doi: 10.48550/arXiv.1901.08644.
- Mohanty, H., Roudsari, A. H., & Lashkari, A. H. (2022). Robust stacking ensemble model for darknet traffic classification under adversarial settings. *Computers & Security*, 120, Article 102830. <https://doi.org/10.1016/j.cose.2022.102830>
- Moustafa, N. (2019a). A Systemic IoT-Fog-Cloud Architecture for Big-Data Analytics and Cyber Security Systems: A Review of Fog Computing (arXiv:1906.01055). arXiv. doi: 10.48550/arXiv.1906.01055.
- Moustafa, N. (2019b). New Generations of Internet of Things Datasets for Cybersecurity Applications based Machine Learning: TON_IoT Datasets.
- Moustafa, N. (2021). A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets. *Sustainable Cities and Society*, 72, Article 102994. <https://doi.org/10.1016/j.scs.2021.102994>
- Moustafa, N., Ahmed, M., & Ahmed, S. (2020). Data Analytics-Enabled Intrusion Detection: Evaluations of ToN_IoT Linux Datasets. In *2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)* (pp. 727–735). <https://doi.org/10.1109/TrustCom50675.2020.00100>
- Moustafa, N., Keshky, M., Debiez, E., & Janicke, H. (2020). Federated TON_IoT Windows Datasets for Evaluating AI-Based Security Applications. In *2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)* (pp. 848–855). <https://doi.org/10.1109/TrustCom50675.2020.00114>
- Moustafa, N., & Slay, J. (2015). UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). *Military Communications and Information Systems Conference (MilCIS), 2015*, 1–6. <https://doi.org/10.1109/MilCIS.2015.7348942>
- Pei, X., Deng, X., Tian, S., Jiang, P., Zhao, Y., & Xue, K. (2024). A Privacy-Preserving Graph Neural Network for Network Intrusion Detection. In *IEEE Transactions on Dependable and Secure Computing*, 1–16. *IEEE Transactions on Dependable and Secure Computing*. <https://doi.org/10.1109/TDSC.2024.3417853>
- Rust-Nguyen, N., Sharma, S., & Stamp, M. (2023). Darknet traffic classification and adversarial attacks using machine learning. *Computers & Security*, 127, Article 103098. <https://doi.org/10.1016/j.cose.2023.103098>
- Safaei Pour, M., Mangino, A., Friday, K., Rathbun, M., Bou-Harb, E., Iqbal, F., Samtani, S., Crichigno, J., & Ghani, N. (2020). On data-driven curation, learning, and analysis for inferring evolving internet-of-Things (IoT) botnets in the wild. *Computers & Security*, 91, Article 101707. <https://doi.org/10.1016/j.cose.2019.101707>
- Searchlight Cyber and Marsh McLennan. (2024, September 23). The Correlation Between Dark Web Exposure and Cybersecurity Risk. <https://slcyber.io/whitepapers-reports/the-correlation-between-dark-web-exposure-and-cybersecurity-risk/>
- Sheikholeslami, S., Meister, M., Wang, T., Payberah, A. H., Vlassov, V., & Dowling, J. (2021). AutoAblation: Automated Parallel Ablation Studies for Deep Learning. In *Proceedings of the 1st Workshop on Machine Learning and Systems* (pp. 55–61). <https://doi.org/10.1145/3437984.3458834>
- Singh, D., Shukla, A., & Sajwan, M. (2021). Deep transfer learning framework for the identification of malicious activities to combat cyberattack. *Future Generation Computer Systems*, 125, 687–697. <https://doi.org/10.1016/j.future.2021.07.015>
- Subasi, A. (2020). *Practical Machine Learning for Data Analysis Using Python*. Academic Press.
- Subasi, A., Algebsani, S., Alghamdi, W., Kremic, E., Almaasrani, J., & Abdulaziz, N. (2021). Intrusion Detection in Smart Healthcare Using Bagging Ensemble Classifier. In A. Badnjivic, & L. Gurbeta Pokvić (Eds.), *CMBBEIH 2021* (pp. 164–171). Springer International Publishing. https://doi.org/10.1007/978-3-030-73909-6_18
- Tazi, F., Shrestha, S., De La Cruz, J., & Das, S. (2022). SoK: An Evaluation of the Secure End User Experience on the Dark Net through Systematic Literature Review. *Journal of Cybersecurity and Privacy*, 2(2), Article 2. <https://doi.org/10.3390/jcp2020018>
- Tran, D.-H., & Park, M. (2024). FN-GNN: A Novel Graph Embedding Approach for Enhancing Graph Neural Networks in Network Intrusion Detection Systems. *Applied Sciences*, 14(16), Article 16. <https://doi.org/10.3390/app14166932>
- Velickovic, P., Cucurull, G., Casanova, A., Romero, A., Lio, P., & Bengio, Y. (2018). Graph Attention. *Networks*. <https://doi.org/10.48550/arXiv.1710.10903>
- Wang, C., & Zhu, H. (2022). Wrongdoing Monitor: A Graph-Based Behavioral Anomaly Detection in Cyber Security. *IEEE Transactions on Information Forensics and Security*, 17, 2703–2718. *IEEE Transactions on Information Forensics and Security*. doi: 10.1109/TIFS.2022.3191493.
- Zhang, P., Zeng, G., Wang, T., & Lu, W. (2024). TinyLlama: An Open-Source Small Language Model (arXiv:2401.02385). arXiv. doi: 10.48550/arXiv.2401.02385.
- Zhang, Z., Hamadi, H. A., Damiani, E., Yeun, C. Y., & Taher, F. (2022). Explainable Artificial Intelligence Applications in Cyber Security: State-of-the-Art in Research. *IEEE Access*, 10, 93104–93139. <https://doi.org/10.1109/ACCESS.2022.3204051>
- Zhu, Y., Tao, J., Wang, H., Yu, L., Luo, Y., Qi, T., Wang, Z., & Xu, Y. (2024). DGNN: Accurate Darknet Application Classification Adopting Attention Graph Neural Network. *IEEE Transactions on Network and Service Management*, 21(2), 1660–1671. *IEEE Transactions on Network and Service Management*. doi: 10.1109/TNSM.2023.3344580.